Lecture 6

Fourier series

Objectives. *An integrable periodic function has Fourier coefficients whether or not its Fourier series converges pointwise. We prove that averaging the partial sums recovers every continuous periodic function uniformly. This gives the Fourier basis of $L^2(\mathbb{T})$ and Parseval's identity. Pointwise convergence requires a separate argument and separate hypotheses.*

By the end of this lecture you will be able to:

- Prove uniform convergence of the Fejér means for continuous periodic functions.
- Expand an L^2 function in the Fourier basis and apply Parseval's identity.
- Relate periodic smoothness to coefficient decay and uniform convergence.
- Prove completeness of the sine basis on a finite interval.

6.1 Coefficients and orthogonal projections

The circle $\mathbb{T} = \mathbb{R}/(2\pi\mathbb{Z})$ identifies points whose difference is an integer multiple of 2π. Functions on $\mathbb{T}$ are 2π-periodic functions. The space $C(\mathbb{T})$ consists of continuous periodic functions. For $1 \le p \le \infty$, we identify $L^p(\mathbb{T})$ with $L^p(-\pi, \pi)$, using periodic representatives. Integration uses $[-\pi, \pi]$ with Lebesgue measure, without a normalising factor. The inner product of $L^2(\mathbb{T})$ is

$$\langle f, g \rangle = \int_{-\pi}^{\pi} \overline{f(x)} g(x) \, \mathrm{d}x.$$

It is antilinear in the first slot. By Theorem 5.9, this space is complete. The functions and coefficients used throughout this lecture are

$$e_n(x) = \frac{e^{\mathrm{i}nx}}{\sqrt{2\pi}}, \qquad c_n(f) = \langle e_n, f \rangle = \frac{1}{\sqrt{2\pi}} \int_{-\pi}^{\pi} e^{-\mathrm{i}nx} f(x) \, \mathrm{d}x, \qquad n \in \mathbb{Z}.$$

The coefficient is defined for every $f \in L^1(\mathbb{T})$. Integration of $e^{\mathrm{i}(n-m)x}$ gives $\langle e_m, e_n \rangle = 0$ when $m \neq n$ and 1 when $m = n$. Thus the family is *orthonormal*.

For $f \in L^2(\mathbb{T})$ define its symmetric partial sum by

$$S_N f = \sum_{|n| \le N} c_n(f) e_n, \qquad N \in \mathbb{N}_0.$$

The difference $f - S_N f$ is orthogonal to every e_n with $|n| \le N$. Expanding the norm therefore gives

$$\|f - S_N f\|_2^2 = \|f\|_2^2 - \sum_{|n| \le N} |c_n(f)|^2. \tag{6.1}$$

Nonnegativity proves *Bessel's inequality*: $\sum_{n \in \mathbb{Z}} |c_n(f)|^2 \le \|f\|_2^2$. It also proves that $S_N f$ is the best approximation in the span of these harmonics. Indeed, for any p in this span, orthogonality gives $\|f - p\|_2^2 = \|f - S_N f\|_2^2 + \|S_N f - p\|_2^2$.

Proposition 6.1 (Riemann–Lebesgue). *For $f \in L^1(\mathbb{T})$, the coefficients satisfy $c_n(f) \to 0$ as $|n| \to \infty$.*

Proof. Write $\tau_h f(x) = f(x-h)$, periodically. A change of variable gives $c_n(\tau_h f) = e^{-inh} c_n(f)$. For $n \neq 0$, take $h = \pi/n$. The coefficient bound then gives

$$2|c_n(f)| = |c_n(f - \tau_{\pi/n} f)| \leq \frac{1}{\sqrt{2\pi}} \left\| f - \tau_{\pi/n} f \right\|_1 \longrightarrow 0.$$

Translation continuity follows from Theorem 5.5 on an interval: extend f periodically to $(-2\pi, 2\pi)$ and set it to zero outside. For $|h| < \pi$, its translated difference on $[-\pi, \pi]$ is exactly the periodic difference above and is bounded by the whole-line norm. ∎

For example, $f = e_2 + 3e_{-1}$ has just two nonzero coefficients, $c_2 = 1$ and $c_{-1} = 3$. Its squared norm is 10. Bessel's inequality becomes an equality once $N \geq 2$. We next prove equality for every L^2 function.

6.2 Positive kernels and averaged sums

The circle convolution has the same unnormalised convention as the whole-line convolution of Lecture 5:

$$(f * g)(x) = \int_{-\pi}^{\pi} f(x-t) g(t) \, \mathrm{d}t.$$

For $f, g \in L^1(\mathbb{T})$ it exists almost everywhere and is integrable, by Tonelli (Theorem 4.1) and periodic change of variable. In particular, $c_n(f * g) = \sqrt{2\pi}\, c_n(f) c_n(g)$, by Fubini from the same standing fact. We will use convolution with continuous kernels, when it exists at every x.

The *Dirichlet kernel* and the *Fejér kernel* are

$$D_N(t) = \sum_{|n| \leq N} e^{int} \quad (N \in \mathbb{N}_0), \qquad F_N(t) = \frac{1}{N} \sum_{m=0}^{N-1} D_m(t), \qquad N \geq 1.$$

Substituting the coefficients into $S_N f$ shows that $S_N f = (2\pi)^{-1} f * D_N$. The *Fejér mean* is the average

$$\sigma_N f = \frac{1}{N} \sum_{m=0}^{N-1} S_m f = \frac{1}{2\pi} f * F_N.$$

Counting the appearances of each frequency in the average gives

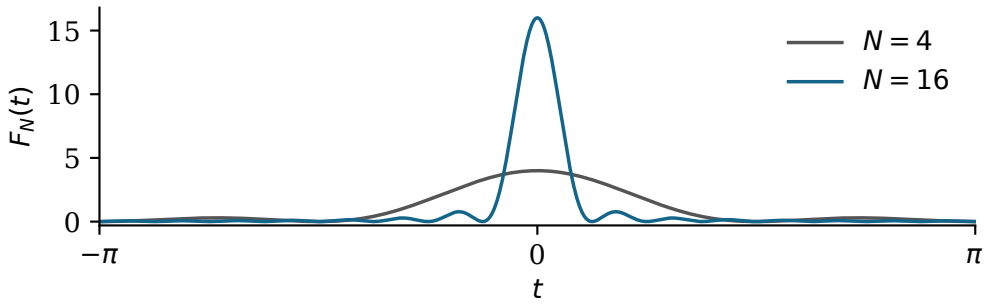
$$F_N(t) = \sum_{|n|<N} \left(1 - \frac{|n|}{N}\right) e^{int} = \frac{1}{N} \left| \sum_{j=0}^{N-1} e^{ijt} \right|^2 = \frac{\sin^2(Nt/2)}{N \sin^2(t/2)}. \tag{6.2}$$

For the middle equality, expand the square. There are $N - |n|$ pairs of indices with difference n. The last equality is the finite geometric-sum formula for $t \notin 2\pi\mathbb{Z}$. The continuous value at $t = 0$ is N. Consequently $F_N \geq 0$ and $\int_{-\pi}^{\pi} F_N = 2\pi$. For every $0 < \delta < \pi$, the mass away from zero satisfies

$$\int_{\delta \leq |t| \leq \pi} F_N(t) \, \mathrm{d}t \leq \frac{2\pi}{N \sin^2(\delta/2)} \longrightarrow 0. \tag{6.3}$$

A family with these positivity, mass and concentration properties is an *good kernel* on $\mathbb{T}$. After division by 2π it has the circle version of the approximate-identity properties.

The graph illustrates concentration while the integral stays fixed.



The Fejér kernels for $N = 4$ and $N = 16$. Both have integral 2π. The formula in (6.3) bounds the mass outside any fixed neighbourhood of zero.

Theorem 6.2 (Fejér). *For $f \in C(\mathbb{T})$, the means $\sigma_N f$ converge uniformly to f.*

Proof. Uniform continuity gives $\delta \in (0, \pi)$ such that $|f(x-t) - f(x)| < \varepsilon$ whenever $|t| < \delta$, for all $x \in \mathbb{T}$. Using the mass and positivity of F_N, split the difference into these small translations and their complement:

$$|\sigma_N f(x) - f(x)| \le \varepsilon + \frac{2\,\|f\|_\infty}{2\pi} \int_{\delta \le |t| \le \pi} F_N(t)\,\mathrm{d}t.$$

The second term tends to zero independently of x, by (6.3). Taking the supremum and then letting $\varepsilon \downarrow 0$ proves the claim. ∎

The kernel argument and uniform approximation theorem are also given in Stein–Shakarchi, *Fourier Analysis*, Chapter 2, §5.2, pp. 53–54. Their coefficient and convolution conventions differ from ours.[1]

6.3 Completeness and Parseval

A *trigonometric polynomial* is a finite linear combination of the functions e_n. Each Fejér mean is such a polynomial, so Fejér's theorem proves their uniform density in $C(\mathbb{T})$.

They are also dense in $L^2(\mathbb{T})$. To see this, M6 (Theorem 5.4) approximates any $L^2(-\pi, \pi)$ function by a continuous function compactly supported in $(-\pi, \pi)$. Its zero values near the endpoints make its periodic extension continuous on $\mathbb{T}$. Uniform approximation of this extension implies L^2 approximation, since $\|h\|_2 \le \sqrt{2\pi}\,\|h\|_\infty$.

An orthonormal family whose finite linear combinations are dense is an *orthonormal basis*. This definition concerns norm convergence, with no claim of pointwise convergence.

Theorem 6.3 (Fourier basis and Parseval). *The family $(e_n)_{n\in\mathbb{Z}}$ is an orthonormal basis of $L^2(\mathbb{T})$. For every $f \in L^2(\mathbb{T})$,*

$$S_N f \longrightarrow f \quad \text{in } L^2(\mathbb{T}), \qquad \sum_{n\in\mathbb{Z}} |c_n(f)|^2 = \|f\|_2^2 .$$

Proof. Given $\varepsilon > 0$, choose a trigonometric polynomial p with $\|f - p\|_2 < \varepsilon$. For all N containing its frequencies, the best-approximation property gives $\|f - S_N f\|_2 \le \|f - p\|_2 < \varepsilon$. Now let $N \to \infty$ in (6.1) to obtain Parseval's identity. ∎

[1] Their coefficient $\widehat{f}(n)$ is $(2\pi)^{-1/2} c_n(f)$. Their circle convolution contains $1/(2\pi)$; ours places this factor before $f * F_N$. Their inner products are linear in the first slot. On the circle, their normalised inner product is $(f, g)_{\mathrm{SS}} = (2\pi)^{-1} \langle g, f \rangle$ (Chapter 3, p. 75).

For example, a function orthogonal to every harmonic is zero in L^2: all its coefficients vanish, so Parseval gives zero norm. Equality here means equality almost everywhere. Changing a value at one point changes neither the coefficients nor the L^2 element.

6.4 Pointwise convergence at jumps

Norm convergence does not determine point values. The following separate result applies to a periodic representative with controlled one-sided behaviour. A *piecewise* C^1 periodic function has a finite partition of one period such that each restriction extends to a C^1 function on the closed subinterval. Adjacent extensions may have different values. Write $f(x+0)$ and $f(x-0)$ for the one-sided limits, interpreted periodically at π and $-\pi$.

Theorem 6.4 (Dirichlet · *sketched*). *For a piecewise C^1 periodic function f and every $x \in \mathbb{T}$,*

$$\lim_{N\to\infty} S_N f(x) = \frac{f(x+0)+f(x-0)}{2}.$$

Proof sketch. The geometric-sum formula gives $D_N(t) = \sin((N+\frac{1}{2})t)/\sin(t/2)$ away from zero. Fix x and put $A = (f(x+0)+f(x-0))/2$. Evenness and the mass $\int D_N = 2\pi$ give

$$S_N f(x) - A = \frac{1}{\pi}\int_0^{\pi} q_x(t)\sin((N+\tfrac{1}{2})t)\,\mathrm{d}t, \qquad q_x(t) = \frac{f(x+t)+f(x-t)-2A}{2\sin(t/2)}.$$

The one-sided C^1 hypothesis makes the numerator $O(t)$ near zero. Thus q_x is bounded there and integrable on $(0,\pi)$. Extend $q_x(t)e^{\mathrm{i}t/2}$ and $q_x(t)e^{-\mathrm{i}t/2}$ by zero to $(-\pi,0)$. Writing the sine as two exponentials reduces the integral to Fourier coefficients of these $L^1(\mathbb{T})$ functions. They tend to zero by Proposition 6.1. This gives the asserted limiting value. The one-sided derivative bounds and the finite partition supply the boundedness assertions used in this sketch. ∎

For example, the periodic extension of $f(x) = x$ on $(-\pi,\pi)$ has a jump at the identified endpoint. Its series converges to x at every interior point and to 0 at π. It cannot converge uniformly to this periodic function: all partial sums are continuous on $\mathbb{T}$, whereas the function has a jump.

6.5 Periodic smoothness and coefficient decay

The notation $C^k(\mathbb{T})$ means a 2π-periodic function with k continuous derivatives on $\mathbb{R}$. Its endpoint values and derivatives through order k match. For $f \in C^1(\mathbb{T})$, integration by parts has no boundary term and gives

$$c_n(f') = \mathrm{i}n c_n(f), \qquad n \in \mathbb{Z}. \tag{6.4}$$

For $n = 0$, this says $\int_{-\pi}^{\pi} f' = 0$ by periodicity. Repeating the identity and applying Riemann–Lebesgue to $f^{(k)}$ proves

$$n^k c_n(f) \longrightarrow 0 \quad \text{as } |n| \to \infty, \qquad f \in C^k(\mathbb{T}), \quad k \in \mathbb{N}_0.$$

Proposition 6.5. *If $f \in C^1(\mathbb{T})$, its Fourier series converges absolutely and uniformly to f.*

Proof. Bessel applied to f' and (6.4) give $\sum_{n\neq 0} n^2|c_n(f)|^2 \le \|f'\|_2^2$. Cauchy–Schwarz therefore gives

$$\sum_{n\neq 0} |c_n(f)| \le \|f'\|_2 \left(\sum_{n\neq 0} \frac{1}{n^2}\right)^{1/2} < \infty.$$

Since $|e_n(x)| = 1/\sqrt{2\pi}$, the Weierstrass M-test (Theorem 1.3) gives absolute and uniform convergence to a continuous function g. Uniform convergence implies L^2 convergence. The same partial sums converge to f in L^2 by Theorem 6.3. Thus $g = f$ almost everywhere. Two continuous functions equal almost everywhere are equal everywhere: a nonzero difference at a point would remain nonzero on an interval of positive measure. ∎

The endpoint condition matters. The function x is smooth on the closed interval $[-\pi, \pi]$, but does not define a continuous function on $\mathbb{T}$. Its boundary term survives integration by parts. This explains its slower coefficient decay and the failure of uniform convergence above.

6.6 The sine basis and a series sum

On an interval $(0, L)$ with $L > 0$, define

$$u_n(x) = \sqrt{\frac{2}{L}} \sin \frac{n\pi x}{L}, \qquad n \ge 1.$$

The sine product identity gives $\langle u_m, u_n\rangle = 0$ for $m \neq n$ and 1 for $m = n$. We prove completeness by odd extension.

Theorem 6.6. *The family $(u_n)_{n\ge 1}$ is an orthonormal basis of $L^2(0, L)$.*

Proof. For $f \in L^2(0, L)$ define $f_{\text{odd}}(x) = f(x)$ on $(0, L)$ and $f_{\text{odd}}(x) = -f(-x)$ on $(-L, 0)$, then extend with period $2L$. The substitution $t = \pi x/L$ transfers Theorem 6.3 to $(-L, L)$, whose orthonormal harmonics are $v_n(x) = (2L)^{-1/2} e^{in\pi x/L}$. Oddness removes the constant and cosine terms. If $b_n = \langle u_n, f\rangle$, direct integration gives

$$\langle v_n, f_{\text{odd}}\rangle = -ib_n, \qquad \langle v_{-n}, f_{\text{odd}}\rangle = ib_n \quad (n \ge 1), \qquad \langle v_0, f_{\text{odd}}\rangle = 0.$$

Pairing the positive and negative terms gives $(-ib_n)v_n + (ib_n)v_{-n} = b_n u_n$ on $(0, L)$. The full Fourier partial sums converge to f_{odd} in $L^2(-L, L)$. Their restrictions therefore converge to f in $L^2(0, L)$. This proves density of the sine span and hence completeness. ∎

Finally take the periodic sawtooth $f(x) = x$ on $(-\pi, \pi)$. Oddness gives $c_0 = 0$. Integration by parts gives, for $n \neq 0$,

$$c_n = \sqrt{2\pi}\, \frac{i(-1)^n}{n}, \qquad \|f\|_2^2 = \int_{-\pi}^{\pi} x^2\, dx = \frac{2\pi^3}{3}.$$

Parseval counts both positive and negative frequencies:

$$\frac{2\pi^3}{3} = 4\pi \sum_{n=1}^{\infty} \frac{1}{n^2}, \qquad \sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}.$$

The sum follows from norm convergence. It does not require pointwise convergence at the sawtooth's jump.

The Fourier and sine basis theorems discharge QM-II's standing fact S2.

Summary.

- Fejér means recover every continuous periodic function uniformly. Their density proves the Fourier basis theorem and Parseval in $L^2(\mathbb{T})$.
- Periodic C^1 smoothness gives absolute and uniform convergence of the ordinary Fourier series.
- The separate Dirichlet statement gives the mean of a jump under piecewise C^1 hypotheses.
- Odd extension gives the complete sine basis on $(0, L)$, for $L > 0$.

References

E. M. Stein and R. Shakarchi, *Fourier Analysis: An Introduction*, Princeton, 2003, Chapter 2, pp. 42–54. N. Jacob and K. P. Evans, *A Course in Analysis IV*, World Scientific, 2018, Chapter 5, pp. 69–71. The Fourier coefficient and convolution conversions are stated in § 6.2.